

1. Introduction

Automatic text recognition in images plays a central role in many computer vision applications, such as administrative document analysis, digital archiving, video surveillance, or the digitization of content captured by mobile devices. Despite significant progress in optical character recognition (OCR) systems, their performance remains highly dependent on the visual quality of the input images. Under real-world conditions, text images are frequently degraded, making recognition partial or incorrect.

Among the most common degradations are motion blur, related to unstable image acquisition or moving objects, low-light conditions inducing noise and contrast loss, as well as artifacts resulting from the scanning of paper documents. These perturbations particularly affect fine text structures, such as character contours and thin strokes, which are essential for reliable recognition. Text thus differs from natural objects by a strong geometric regularity and an increased sensitivity to local distortions.

The impact of these degradations on OCR performance is significant. Even moderate alterations can lead to a substantial increase in recognition errors. In sensitive contexts, particularly administrative or legal ones, these errors compromise the extraction of critical information and the reliability of automated systems. Improving text readability prior to recognition therefore constitutes a major challenge for strengthening the robustness of document analysis pipelines.

In this context, image restoration approaches, and in particular deblurring techniques, appear as a key step to indirectly improve OCR performance. The objective is not only to obtain a visually more pleasant image, but to restore relevant textual structures in a way that preserves their interpretability by recognition systems.

2. Classical methods and pre-deep learning approaches

Before the rise of deep learning, blurred image restoration mainly relied on analytical and statistical approaches based on explicit degradation models. These methods aimed to invert the blurring process using simplifying assumptions regarding the blur kernel and the statistical properties of the original image. Although they provided an important theoretical foundation, their performance remains limited in complex scenarios.

Wiener filtering models restoration as a linear estimation problem in the presence of additive noise, assuming prior knowledge of the signal and noise spectra. In practice, this assumption is rarely satisfied, and the method tends to smooth high frequencies, resulting in a loss of fine details. In the case of text, these high frequencies correspond precisely to character contours and strokes, which severely limits readability.

Blind deconvolution methods were subsequently proposed to address situations where the blur kernel is unknown. They jointly estimate the latent image and the kernel from the observed image, relying on assumptions such as gradient sparsity. However, these assumptions are often violated in real text images, leading to artifacts, halos, or local distortions.

Other approaches rely on explicit priors, such as total variation regularization or edge preservation. Although they may improve overall sharpness, they struggle to recover the structural complexity of heavily degraded text. The regularization required for inversion stability frequently leads to over-smoothing, which removes discriminative details necessary for recognition.

In summary, classical methods remain insufficient for effectively restoring text images acquired under real-world conditions. Their dependence on simplifying assumptions and their limited adaptability motivated the development of deep learning based approaches.

3. Deep learning for image restoration

The introduction of deep neural networks has profoundly transformed image restoration, particularly for deblurring. Unlike classical methods, these approaches directly learn the nonlinear relationship between a degraded image and its restored counterpart from large amounts of data, which has led to significant gains in quality and robustness.

3.1 CNN-based approaches

Early deep learning based deblurring methods rely on convolutional networks trained in a supervised manner by minimizing reconstruction losses such as L1 or L2. These models outperform classical approaches on complex blur patterns and combined noise, while remaining relatively computationally efficient. Nevertheless, pixel-wise optimization often favors over-smoothed reconstructions, which is problematic for text readability.

3.2 U-Net architectures and variants

U-Net-type architectures, based on an encoder-decoder structure with skip connections, preserve low-level spatial information while integrating more abstract representations. They offer a good compromise between representational capacity and training stability, and are particularly well suited for the local restoration of textual structures. However, their generalization remains limited when degradation conditions differ from the training data.

3.3 GAN-based methods

Generative adversarial networks, such as DeblurGAN and DeblurGAN-v2, were introduced to limit the over-smoothing observed in purely convolutional approaches.

By combining adversarial loss with reconstruction losses, these models significantly improve perceptual sharpness and contrast in restored images. However, GAN training remains unstable and may introduce artifacts or hallucinations, which can harm the structural fidelity required for reliable OCR recognition. ([DeblurGAN: Blind Motion Deblurring Using Conditional Adversarial Networks](#) / [DeblurGAN-v2: Deblurring \(Orders-of-Magnitude\) Faster and Better](#))

3.4 Multi-scale and recurrent networks

Multi-scale and recurrent architectures, such as Scale-Recurrent Networks (SRN) and MPRNet, progressively process degradations across different resolutions. This strategy allows global distortions to be corrected first before refining local details, which is particularly effective for complex and non-uniform blur. These models achieve state-of-the-art performance on several deblurring benchmarks, at the cost of increased architectural complexity and computational overhead, which may limit real-time deployment. ([Scale-recurrent Network for Deep Image Deblurring](#) / [Multi-Stage Progressive Image Restoration](#))

Critical summary.

Deep learning based methods largely outperform classical approaches, but involve trade-offs between visual quality, computational cost, and generalization. In text-related scenarios, these trade-offs are further amplified by the need to preserve fine structures essential for recognition.

4. Text-oriented restoration and OCR-aware methods

Text exhibits specific structural characteristics that make its restoration particularly challenging. Composed of thin strokes and repetitive patterns, characters are highly sensitive to local degradations, even mild ones, which may be sufficient to compromise automatic recognition. A visually acceptable image therefore does not necessarily imply an improvement in OCR performance.

Several studies have highlighted the limitations of generic restoration methods when applied to text images. Optimizing pixel-wise or perceptual metrics may produce subtle artifacts that are tolerable for a human observer but detrimental to recognition systems.

To address these limitations, so-called OCR-aware approaches explicitly integrate recognition-related constraints into the restoration process. This includes the use of losses based on character error rates or signals derived from pre-trained OCR networks. Another direction consists in jointly optimizing restoration and recognition within multi-task architectures, where the OCR module acts as semantic supervision.

These methods rely on specialized datasets, such as ICDAR for real annotated text images, SynthText for controlled synthetic data generation, and RealBlur for realistic

blur acquisition. They emphasize the importance of multimodal evaluation and integrated pipelines, in which restoration explicitly aims to improve text recognition.

However, few works propose a systematic comparative analysis of different restoration architectures specifically applied to text images, while jointly evaluating the visual quality of restored images and OCR recognition performance.

5. Evaluation metrics and limitations

The evaluation of restoration methods traditionally relies on objective metrics such as PSNR and SSIM, which measure similarity between the restored image and a reference. Although standardized, these metrics exhibit important limitations for images containing text.

PSNR favors smoothed reconstructions that minimize average error, while SSIM remains insufficiently sensitive to local defects affecting thin character strokes. High scores therefore do not guarantee improved OCR performance. To overcome this limitation, recognition-oriented metrics such as Character Error Rate and Word Error Rate are used to directly measure the impact of restoration on text recognition.

Qualitative evaluation nevertheless remains essential to identify artifacts or structural inconsistencies not captured by numerical scores. These observations motivate a multi-criteria evaluation combining classical metrics, OCR indicators, and visual analysis.

6. Project positioning

The project is positioned within the continuity of recent deep learning based image restoration approaches, while specifically targeting images containing text. It relies on representative state-of-the-art architectures, such as U-Net models, multi-scale recurrent networks, and adversarial models, to evaluate their performance in a text-centric context.

The originality of the project lies in its comparative and application-oriented approach. Rather than proposing a new architecture, it aims to systematically analyze the trade-offs between different model families for restoring degraded text. This analysis is complemented by a multimodal evaluation combining visual quality and OCR performance, in order to align restoration objectives with application requirements.

By integrating qualitative and quantitative analyses across different types of degradations, the project positions itself as a coherent study at the intersection of image restoration and text recognition, fully aligned with the expectations of the Machine Vision course.